



A Novel Multi-focus Fusion Network for Retinal Microsurgery

Xinyi Zhou, Louying Hao, Qiushi Nie, Yingquan Zhou, Lihui Wang, Yan Hu*, Jiang Liu



Southern University of Science and Technology
Guangdong Academy of Sciences, Guangzhou



IEEE ISBI 2022
International Symposium on
Biomedical Imaging

Introduction

The retina is a layer of tissue in the back of the eyeball with only about 300-micrometer thickness. Retinal microsurgery, widely adopted for diseases such as macular holes and branch\central retinal vein occlusion, requires extremely high precision. However, due to the limited depth of field (DOF) of the high magnification microscope and the curvature of eyeballs, there are defocused regions in ophthalmic microscopy images. Therefore, as shown in Fig.1, it is more difficult to locate the exact vessel position to inject and supervise the blood flow during retinal endovascular surgery.



Fig.1 Ophthalmic microscopy image in retinal endovascular surgery

Multiple approaches have emerged to obtain all-in-focus images, with multi-focus image fusion the most widely used one. Methods were developed based on transform domain, spatial domain, image pixel operations, and deep neural networks for natural scenes. Failing to address the problem of focus boundaries and contextual information extraction, previous methods did not achieve satisfactory results. In this paper, we propose a novel deep-learning-based multi-focus image fusion model, which can reconstruct all-in-focus images by fusing a stack of reginal out-of-focus ophthalmic microscope images.

Methodology

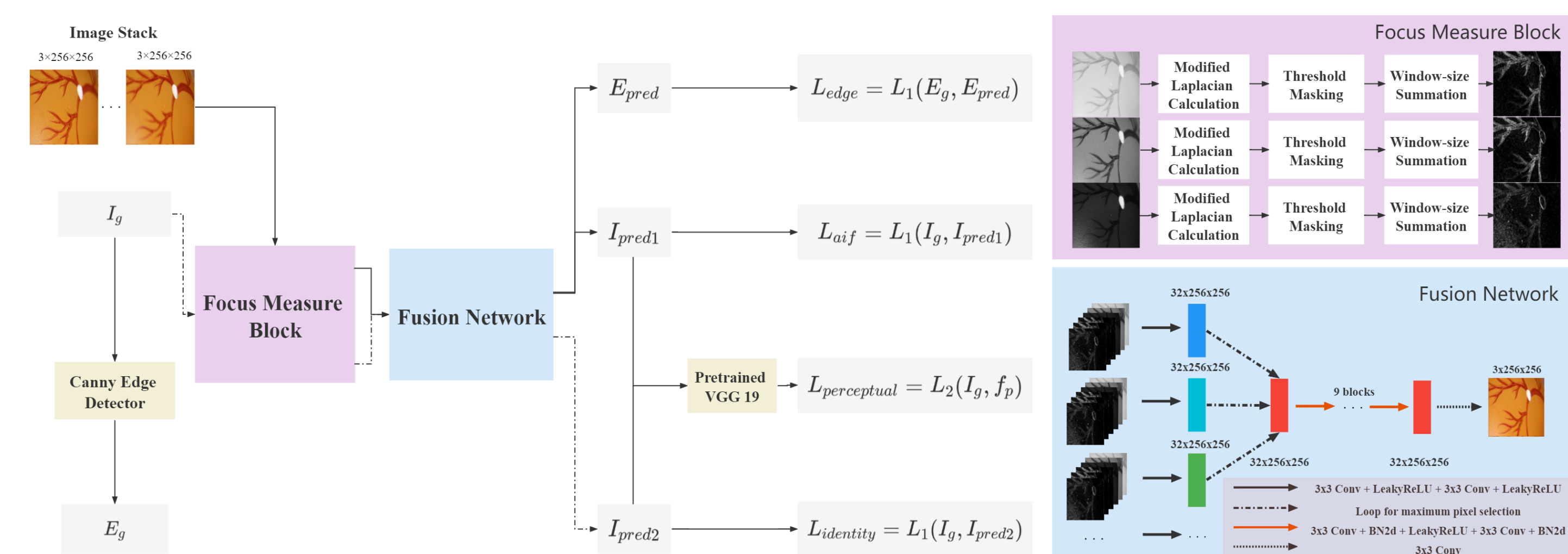


Fig.2 The general framework of the model, with Focus Measure Block and Fusion Network depicted in detail.

Focus Measure Block (FMB)

A stack of images with different defocused areas is input into FMB to measure their focus degree. It adopts the three steps of sum-modified-Laplacian shown in Eqs.(1) and (2) to calculate the focus measure from every channel of the input image. $\nabla_{ML}^2 I(x, y)$ stands for the discrete approximation to the modified Laplacian of each pixel at (x, y) in the one channel matrix I . FMB represents the output focus measure at (x, y) as the sum of modified Laplacian larger than a threshold value T in a window size of N .

$$\nabla_{ML}^2 I(x, y) = |2I(x, y) - I(x - step, y) - I(x + step, y)| + |2I(x, y) - I(x, y - step) - I(x, y + step)| \quad (1)$$

$$FMB = \sum_{i=x-N}^{i=x+N} \sum_{j=y-N}^{j=y+N} \nabla_{ML}^2 I(i, j), \text{ for } \nabla_{ML}^2 I(x, y) \geq T \quad (2)$$

Fusion Network (FN)

Each image from the stack is input into FN concatenated with three-dimensional focus measure matrices. FN can be divided into four modules. The first module does sampling on inputs and stores information in feature maps. Then, the second module selects the maximum values from those feature maps and fuses them together. Next, nine blocks of the third module refine the output features. At last, the optimized features are reconstructed into an all-in-focus image with the fourth network module.

Loss Function

As shown in Eqs.(3), the loss function contains four loss items, L_{aif} , $L_{perceptual}$, L_{edge} , and $L_{identity}$. L_{aif} calculates the L_1 loss between the ground-truths and the output all-in-focus images. $L_{perceptual}$ denotes the perceptual loss between the ground-truths and feature maps from a pretrained CNN model. L_{aif} and $L_{perceptual}$ are utilized instead of MSE loss to avoid the boundary smoothing effect of L_2 -norms. To maintain texture details such as blood vessels in retinal images, L_{edge} is introduced by calculating the L_1 loss between the edge maps of the ground-truths and outputs. Furthermore, the identity loss $L_{identity}$ is considered for the model to learn identical color mapping.

$$L = L_{aif} + L_{perceptual} + L_{edge} + L_{identity} \quad (3)$$

Experiments and Results

Dataset

Our in-house dataset containing 3718 microscope fundus images rescaled to 256x256 with a fake eye model. This dataset is collected with an EDER Surgical Microscope SM2000J under common ophthalmic clinical magnifications of 10, 16, and 20. Each magnification class contains five image stacks for training and one stack for testing. A well-performed traditional fusion algorithm is adopted to generate ground-truth all-in-focus images.

Baseline

In this paper, we proposed the focus measure block (FMB) and loss function based on the fusion network (FN). Thus, the performance of FN is considered the baseline.

Performance Metrics

Structural Similarity (SSIM) and Peak Signal-to-Noise Ratio (PSNR) are used to measure the quality of the reconstructed all-in-focus images.

Ablation Study Results

To prove the effectivity of our FMB, which calculates focus measure for each channel (named as FMB3), we also gave out the experiment results of focus measure block for the gray channel of images, short as FMB1.

Models	FN	FN+FMB1	FN+FMB3	FN+FMB3+ Edge loss	FN+FMB3+ Identity loss	Ours
SSIM	0.8942	0.8860	0.9071	0.9079	0.9193	0.9230
PSNR	30.64	29.86	30.80	31.30	30.82	32.10

Comparison Experiments Results

Models	Wang	ASR	CBF	IFCNN	U2Fusion	AiFNet	Ours
SSIM	0.6782	0.8054	0.7180	0.8583	0.7620	0.9054	0.9230
PSNR	25.04	29.26	26.43	29.35	25.64	31.91	32.10

Conclusion

In this paper, we proposed a multi-focus image fusion model to generate all-in-focus images for ophthalmic microsurgery, which provides the surgeons with a clear view of target tissues. A focus measure block was proposed to obtain the focus areas, and then a fusion network fused them to produce all-in-focus images. Evaluated on our collected database, the ablation and comparison experiments proved the effectiveness of the proposed modules and loss function. In the future, we would like to extend our algorithm to achieve advanced performance on real optic microsurgery images and videos.



(a) Image stack with different defocused area.



(b) Wang (c) ASR (d) CBF (e) IFCNN (f) U2Fusion (g) AiFNet (h) Ours (i) GT
Fig.3 Qualitative comparison of our method with 7 methods and ground truths.

Fusion Application

The limited field of view under the microscope makes it difficult to get a panoramic view of the fundus during ophthalmic surgery. Even with image stitching algorithms, partially blurred microscope images cannot give satisfactory results. With the help of our model and generated all-in-focus images, stitching outputs are more satisfying as shown in Fig.4.

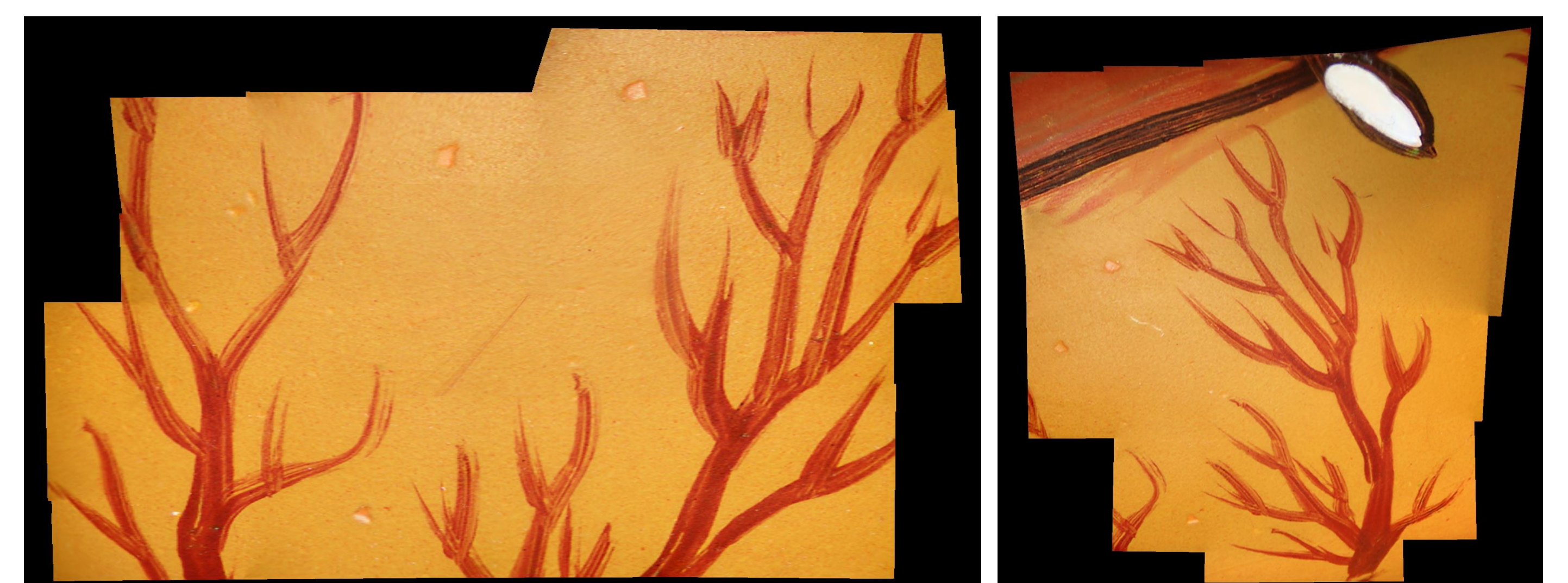


Fig.4 Panorama image results stitched from generated all-in-focus images.